

Pedestrians Movement on Crossings

Final Report

Agent-based simulation of crowd pressure at bottlenecks and crossings

Course: Agent Systems

Team members:

Kamil Barszczak, Radosław Barszczak, Szymon Klempert

June 2026

1. Problem description

With the growing number of mass events and increasing urban density, understanding and predicting the movement of pedestrian crowds has become a safety-critical problem. When many people try to pass through a constrained space - a doorway, a corridor, or a crossing where several streams meet - local density can rise to dangerous levels. The physical force transmitted through a tightly packed crowd, known as crowd pressure, is the direct cause of crowd disasters: people are no longer in control of their own motion and are pushed by the bodies around them.

The goal of this project is to build an agent-based simulator that reproduces this phenomenon from the bottom up. Each pedestrian is modelled as an autonomous agent following simple local rules; no central controller coordinates the crowd. We then measure how the emergent, collective quantities - crowd pressure, walking speed, evacuation time and the fraction of agents that successfully pass - depend on the size of the crowd, the geometry of the environment and the behavioural rules used. The central question is: under which conditions does an orderly flow degrade into a jammed, high-pressure state?

Concretely, the simulator targets three representative scenarios: (i) two groups passing in counter-flow through a single central constriction, (ii) four groups meeting at a crossing from the four cardinal directions, and (iii) two groups crossing through a single narrow gap in a dividing wall. These scenarios let us study both bottleneck congestion and the mixing of opposing pedestrian streams.

2. State of the art

2.1 The Social Force Model

The Social Force Model (SFM), proposed by Helbing and Molnar in 1998, is one of the most widely studied models of pedestrian dynamics. It is a continuous model in which pedestrians are subject to social forces representing their internal motivations. Each pedestrian is influenced by three categories of forces: a driving force toward their goal at a preferred speed, repulsive (and occasionally attractive) forces from other pedestrians, and repulsive forces from walls and obstacles. The model reproduces microscopic interactions very realistically, but its computational cost grows quadratically with the number of agents, which limits scalability.

A key property of the SFM is that it can reproduce the "faster-is-slower" effect: in stressful situations interpersonal distance shrinks, increasing flow but also the chance of physical contact. In the extreme, mutual blocking between agents prevents passage entirely - the "freezing by heating" phenomenon, where a crowd that pushes too hard slows itself down.

2.2 The behavioural heuristics model

An important extension is the behavioural-heuristics model of Moussaïd, Helbing and Theraulaz ("How simple rules determine pedestrian behavior and crowd disasters"). Instead of relying purely on physics-based forces, it governs navigation with two heuristics. The direction heuristic selects the desired heading by minimising a distance function that balances progress toward the goal against the distance to the first obstacle in each candidate direction:

$$d(\alpha) = d_{max}^2 + f(\alpha)^2 - 2 d_{max} f(\alpha) \cos(\alpha_0 - \alpha)$$

The distance heuristic adapts the desired speed so that the time-to-collision stays above the relaxation time τ , giving the desired speed and the resulting acceleration toward it:

$$v_{des} = \min(v_i^0, d_h/\tau) \quad a_i = (v_{des} - v_i)/\tau$$

In dense situations the model adds physical contact forces between agents and against walls, proportional to the body overlap; the full acceleration combines the driving term with these contact forces:

$$a_i = \frac{v_{des} - v_i}{\tau} + \sum_j \frac{f_{ij}}{m_i} + \sum_W \frac{f_{iW}}{m_i}$$

2.3 AI-based approaches

Recent work applies machine learning to pedestrian modelling. Social GAN (Gupta et al., 2018) couples a sequence-to-sequence recurrent network with a generative adversarial network to predict multimodal, socially acceptable trajectories. Deep Social Force (Kreiss, 2021) makes the classical SFM differentiable, replacing fixed interaction potentials with neural function approximators; this removes the unnatural head-on locking of the original model and learns region-specific avoidance biases (e.g. passing on the right). These hybrid approaches keep the interpretable structure of the SFM while leveraging data.

3. Definition of agents

In our system an agent represents a single pedestrian navigating a shared physical environment. Each agent is autonomous: it perceives its local surroundings, decides its movement using the behavioural heuristics, and interacts with others and with obstacles through physical and social forces. There is no central controller - the collective behaviour of the crowd emerges from individual rules.

3.1 Agent attributes

- Mass m (60-90 kg), which also sets the body radius $r = m / 50$.
- Comfortable speed v_0 (1.5-2.0), the preferred walking speed when unobstructed.
- Vision angle ϕ (1.2-1.3 rad), the field of view scanned for obstacles.
- Maximum vision distance d_{max} (100-105), the farthest detection range.
- Relaxation time τ (0.45-0.55 s), how quickly velocity adapts to the desired one.
- Desired position - the destination point, and a group colour identifying its stream.

Each agent also keeps dynamic state updated every step: position, velocity, desired velocity, and the instantaneous crowd pressure exerted on it. Position and velocity use a double-buffering scheme so that agents can be updated concurrently without read-write conflicts.

3.2 Agent types and groups

All agents share the same behavioural model; they are differentiated by parameterisation and group membership. A group is defined by a common spawn area and destination and is drawn in one colour. Within a group, mass, comfortable speed, vision and relaxation time are sampled from the ranges above, producing realistic heterogeneity. The same mechanism could express richer roles (a child with reduced speed, a police officer with different force parameters) purely by changing the initialisation parameters.

3.3 Interactions and expected emergent behaviour

Agents do not exchange explicit messages; all interaction is implicit and mediated by the shared environment. Three interaction types are modelled: agent-agent repulsion when body radii overlap, agent-wall repulsion near walls, and heuristic-based proactive avoidance before contact occurs. From these local rules we expect several collective

phenomena to emerge: lane formation in counter-flow, arching congestion before bottlenecks, group mixing and separation at crossings, and - under excessive pressure - freezing by heating.

4. The simulator application

The simulator is implemented in Java 17 with a JavaFX graphical interface. The design is deliberately modular so that heuristics, the computing engine, the physical model and the map/agent layout can each be swapped without touching the application core.

4.1 Architecture

- Physical model (SocialForcePhysicalModel): turns the social and contact forces into per-step velocity and position updates, and records the crowd pressure on each agent.
- Heuristics (Direction, Distance): set each agent's desired velocity; they can be toggled on/off at runtime.
- Computing engine (single-threaded or thread-pool based): evaluates agent updates, exploiting the double-buffered state for safe parallelism.
- Initializers: procedurally build the wall layout (board) and the agent groups; eight maps are provided, from a single bottleneck to four-way crossings.

4.2 Graphical interface and crowd-pressure visualisation

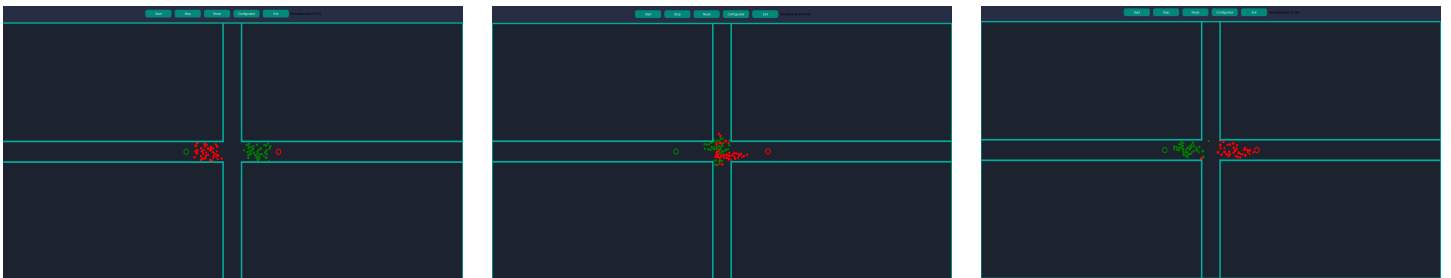
A configuration window lets the user set the number of agents, the force scale coefficient, the destination radius, the simulation time step, the computing engine and the map. The simulation window then animates the crowd in real time with start / stop / reset controls and a running clock. Each agent is drawn in its group colour when calm and is interpolated toward red as the crowd pressure on it rises, so high-pressure regions are immediately visible. Destination areas are drawn as rings and walls as solid lines.

4.3 Data collection

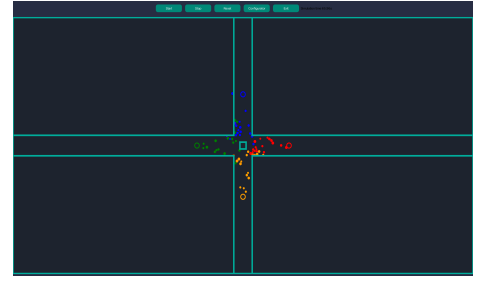
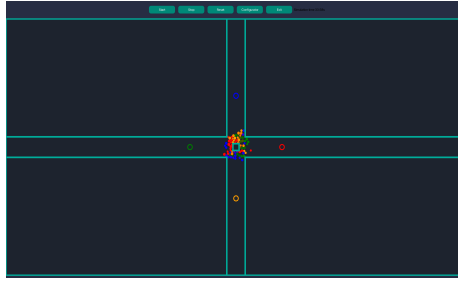
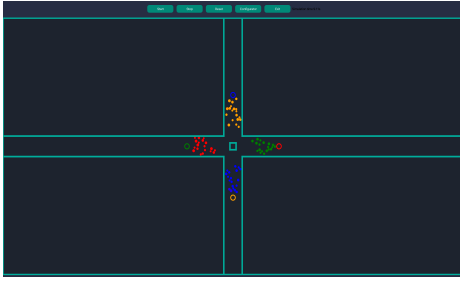
For quantitative analysis the simulator can record, at every step, a CSV row per agent containing its position, velocity, speed, crowd pressure and stopped flag. A headless batch runner sweeps the parameter space (map, heuristics, crowd size) over several repetitions without the GUI, writing a per-run summary plus detailed per-step traces. All figures in Section 5 are produced from these files.

4.4 Example execution sessions

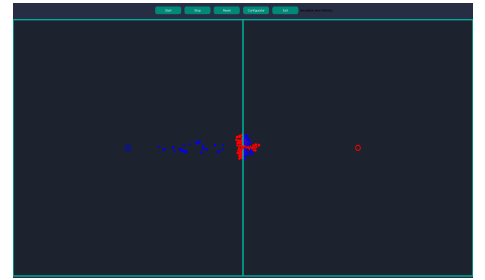
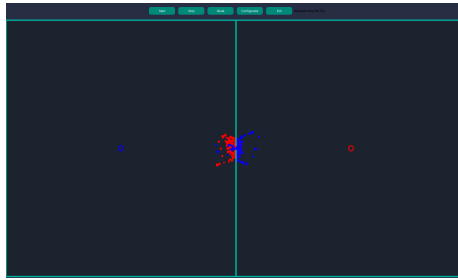
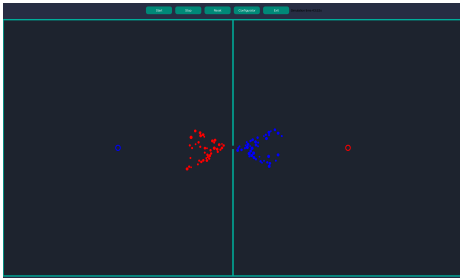
The three figures below show representative runs. In each, time advances left to right; agents fade from their group colour to red as local pressure grows.



Session 1 - two groups (red, green) in counter-flow through the central constriction.



Session 2 - four groups crossing; the streams intermix and build pressure at the centre.



Session 3 - two groups (red, blue) through a single narrow gap in a dividing wall.

5. Analysis of results

5.1 Experimental setup

We ran a parameter sweep with the headless batch runner. Two maps were studied: Map7, a counter-flow bottleneck where two groups cross through a central opening, and Map1, an open corridor with a single dividing wall. For each map we compared two behavioural regimes - heuristics off (raw social forces only) and heuristics on (direction + distance) - across crowd sizes of 25, 50, 75, 100 and 150 agents, with three repetitions per configuration (60 runs total). The force scale coefficient (1000), destination radius (3) and time step (50 ms) were held fixed, and each run was capped at 125 s of simulated time. For every run we recorded the mean and maximum crowd pressure, the mean walking speed, the evacuated fraction and, when all agents cleared, the full-evacuation time.

5.2 Crowd pressure grows super-linearly with crowd size

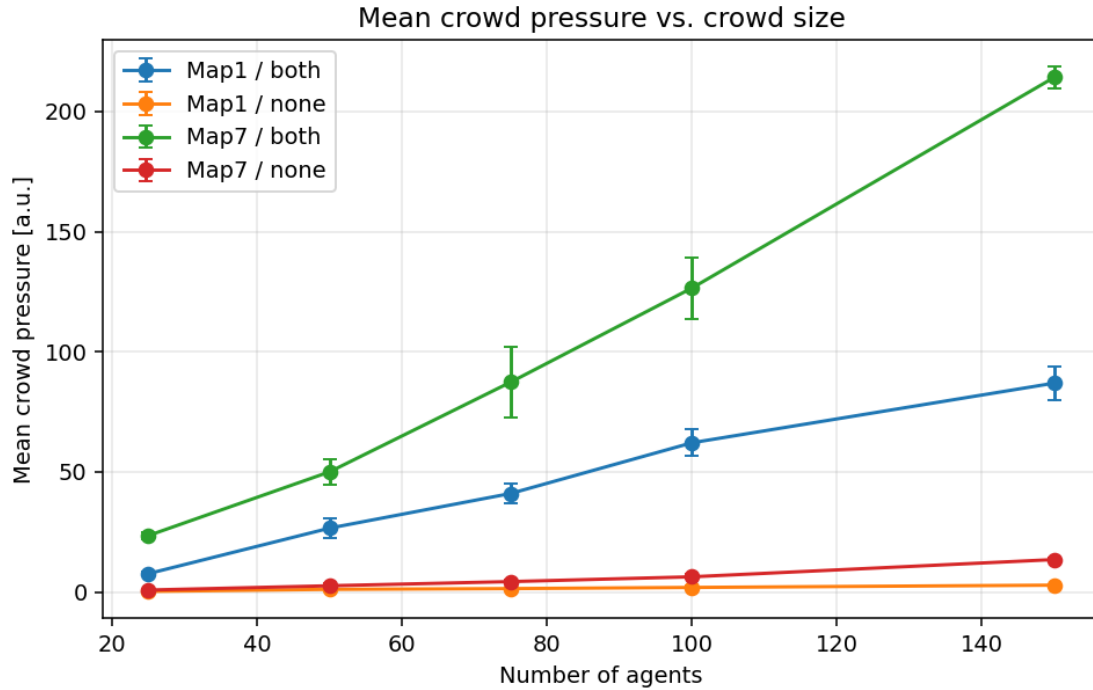


Figure 1. Mean crowd pressure versus number of agents (error bars: std over repetitions).

The mean pressure rises faster than linearly with the number of agents. On the Map7 bottleneck with heuristics it grows from about 24 at 25 agents to about 214 at 150 agents - a roughly nine-fold increase for a six-fold increase in crowd size. This is expected: pressure accumulates from pairwise body overlaps, and both the number of simultaneous contacts per agent and the depth of each overlap increase as the crowd densifies, so their product grows super-linearly. The practical implication is that doubling the crowd more than doubles the force individuals must withstand, which is precisely why bottlenecks become dangerous well before they look completely full.

Pressure is markedly higher with heuristics enabled than without. This may seem counter-intuitive, but it is the correct behaviour: without heuristics agents do not actively steer toward their goal, so they barely enter the constriction and never build up against each other (see 5.3). With heuristics they purposefully push into the bottleneck, which is exactly the situation that generates real crowd pressure.

5.3 Heuristics are essential for navigation

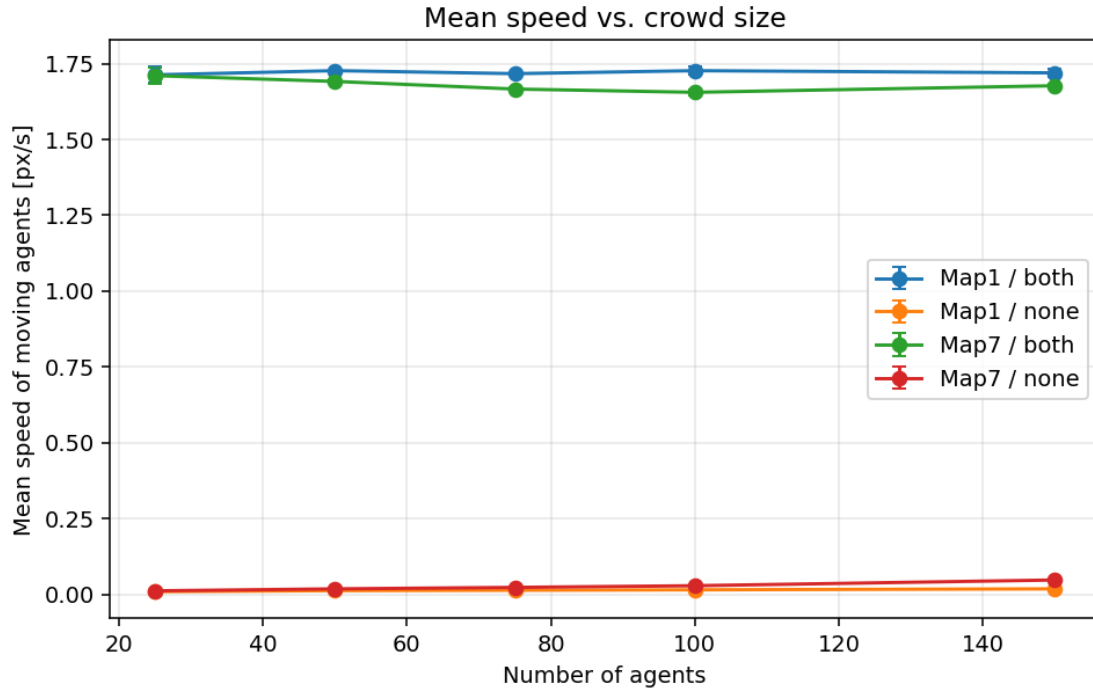


Figure 2. Mean speed of moving agents versus crowd size, with and without heuristics.

Without heuristics the mean speed stays near zero (about 0.01-0.05) and no agent ever reaches its destination: the raw social forces alone keep agents apart but provide no purposeful navigation around the wall toward the goal. With heuristics enabled agents move at close to their comfortable speed (about 1.65-1.73). This confirms that the behavioural heuristics - not the contact forces - are what convert the model into goal-directed pedestrian motion. It also shows a mild but consistent slowdown as the crowd grows (1.71 at 25 agents down to 1.66 at 100 on Map7), an early sign of congestion that becomes decisive in the evacuation metrics below.

5.4 Evacuation degrades as the crowd grows

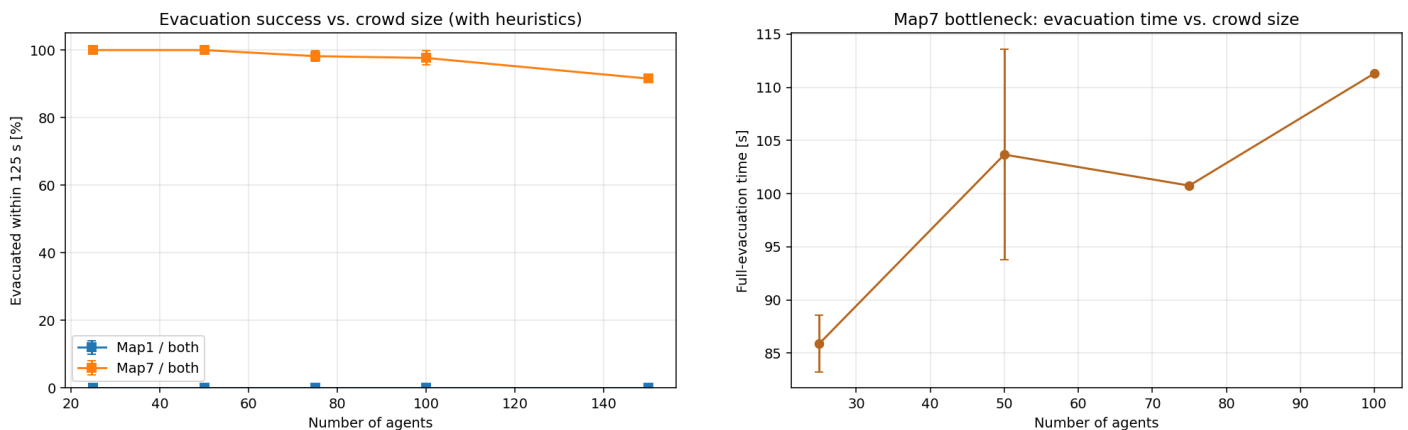


Figure 3. Left: fraction evacuated within 125 s. Right: full-evacuation time vs crowd size (Map7).

On the Map7 bottleneck every agent clears the opening at 25 and 50 agents, but the evacuated fraction then falls - to about 98% at 75 agents, 97% at 100 and only ~92% at 150. At the same time the full-evacuation time rises with crowd size. Together these are the signature of a capacity-limited bottleneck: beyond a critical crowd size the opening can no longer clear everyone within the time window, and a residual jammed cluster remains. This is the macroscopic counterpart of the "faster-is-slower" and "freezing-by-heating" effects described in the literature - more agents pressing into the same gap reduce, rather than increase, the effective throughput.

5.5 Pressure is localised at the bottleneck

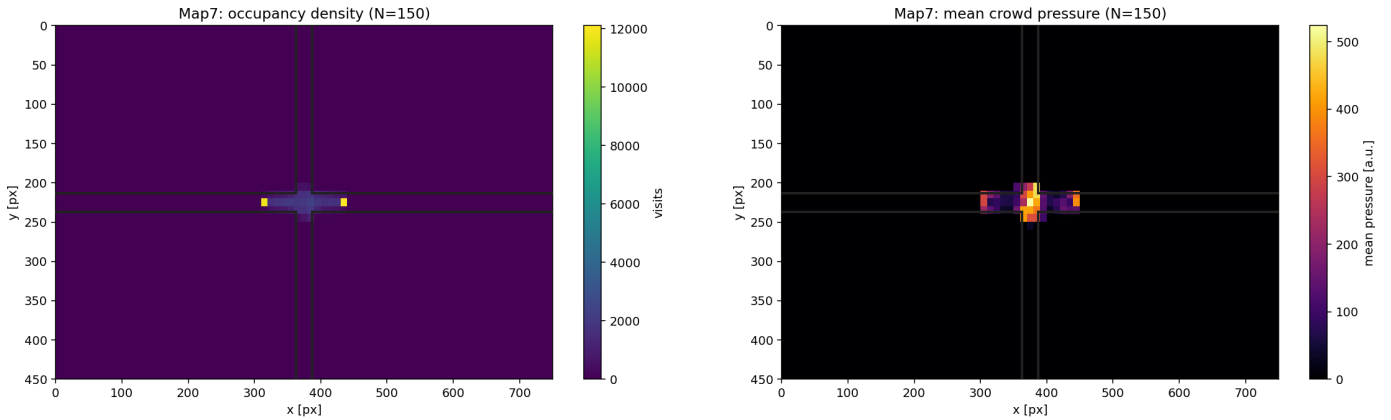


Figure 4. Map7, 150 agents: time-averaged occupancy (left) and mean crowd pressure (right).

The spatial maps confirm where the danger concentrates. Occupancy is spread along the approach corridors, but crowd pressure is sharply localised at the central opening, reaching values far above the surrounding area. In other words, an agent can be in a moderately dense region and feel little force, yet experience extreme pressure the moment it enters the constriction. This matches real observations that crowd disasters occur at specific pinch points rather than uniformly across a venue, and it validates our per-agent pressure metric as a meaningful indicator.

5.6 Lane formation and stream mixing

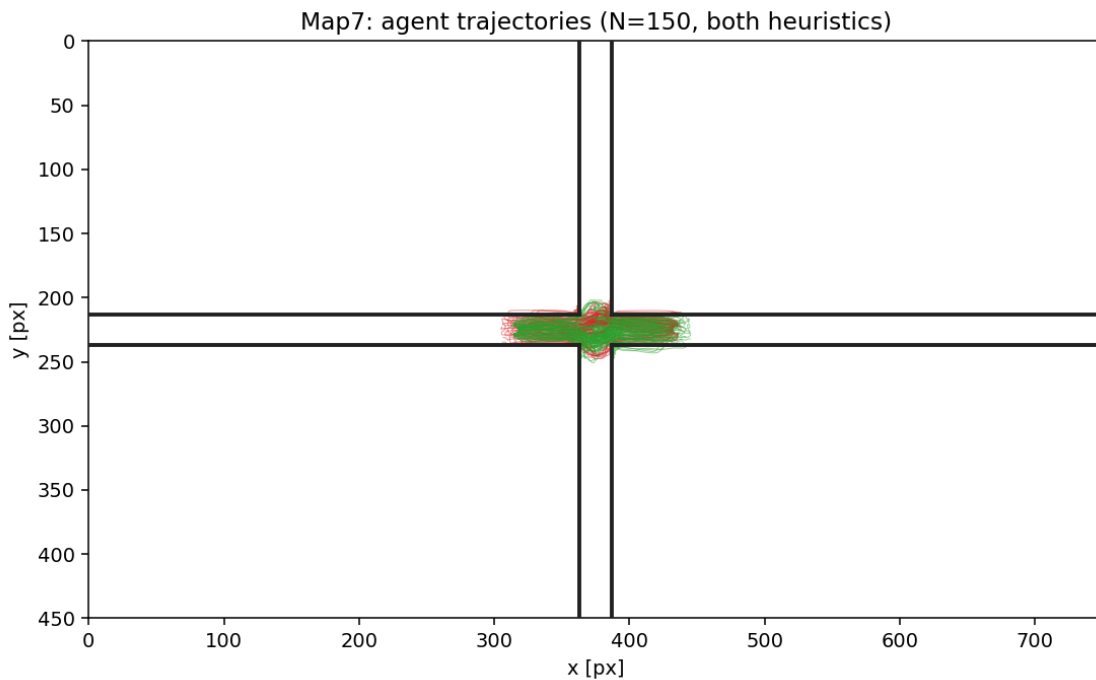


Figure 5. Map7, 150 agents: trajectories coloured by group (red and green counter-flows).

The trajectory bundle shows the two opposing groups converging on the same opening and intermixing tightly before separating on the far side. The dense overlap of red and green paths at the centre is the geometric origin of the pressure peak in Figure 4: counter-flowing agents must negotiate the same few square pixels of free space, repeatedly blocking one another - the micro-mechanism behind the reduced throughput.

5.7 Speed-density relation (fundamental diagram)

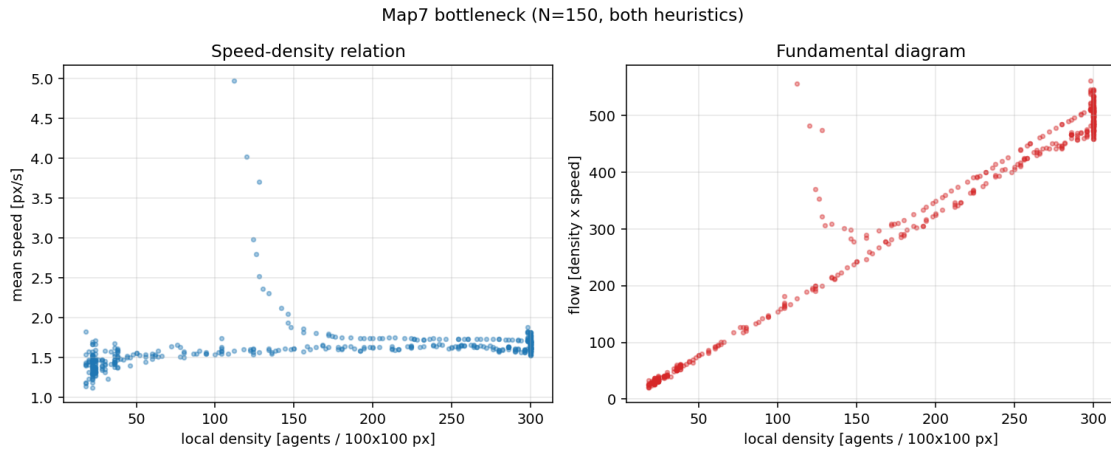


Figure 6. Local speed-density relation and flow measured in a window around the bottleneck.

Measuring local density and speed inside a window around the opening reproduces the classical fundamental diagram of pedestrian traffic: as local density increases, mean speed decreases, and the flow (density $\times$ speed) saturates rather than growing without bound. The fact that an emergent, well-known traffic relationship arises from our simple per-agent rules is a strong validation that the model captures the right physics, not just plausible-looking motion.

5.8 Temporal dynamics of a run

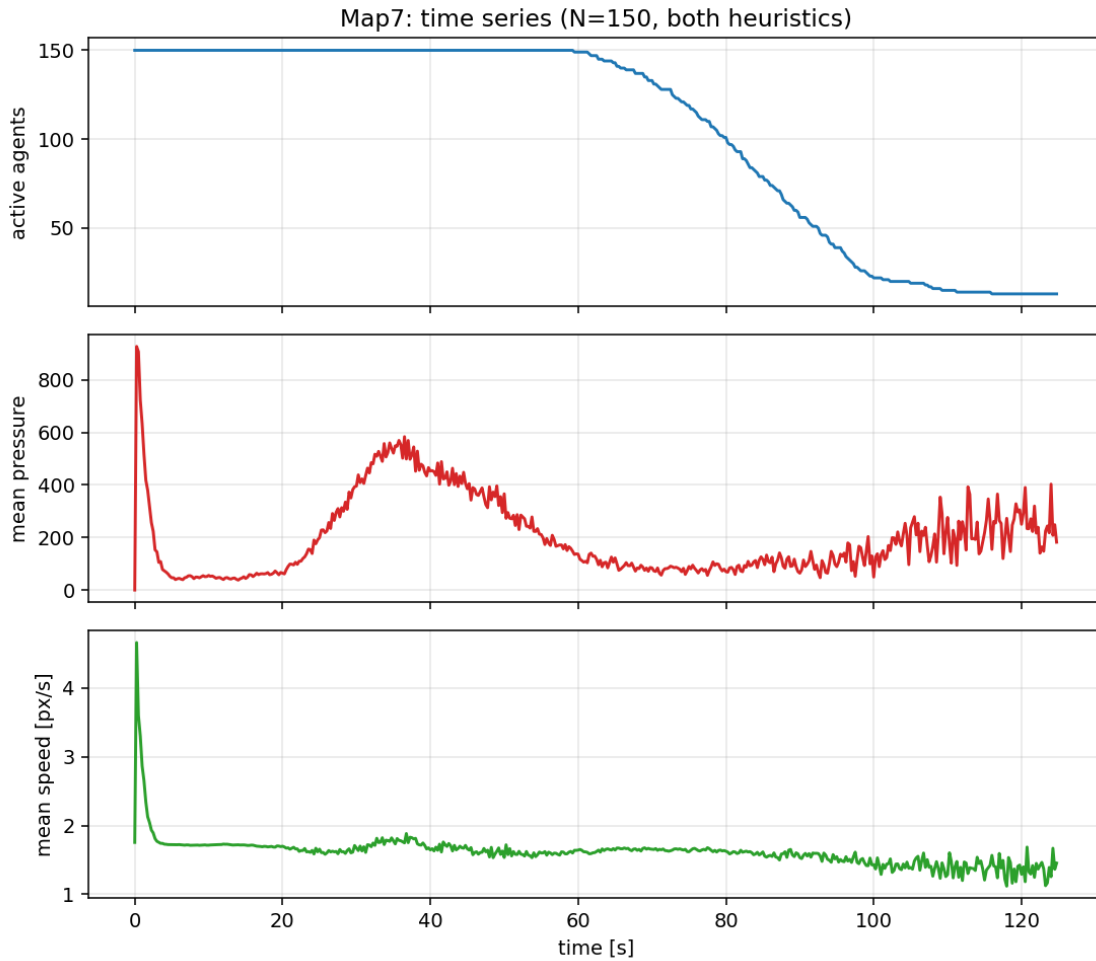


Figure 7. Map7, 150 agents: active agents, mean pressure and mean speed over time.

The time series tells the story of a single run. The number of active agents falls steadily as agents clear the opening; mean pressure rises to a peak while the crowd is packed against the constriction and then decays as the queue drains; mean speed is depressed during the congested phase and recovers toward the end. The peak-pressure phase is exactly the interval during which a real crowd would be at greatest risk.

6. Conclusions and further research

6.1 Conclusions

We built a fully agent-based pedestrian simulator combining the Social Force Model with behavioural heuristics, instrumented it to record per-agent data, and analysed a 60-run parameter sweep. The main findings are: (1) crowd pressure grows super-linearly with crowd size, so risk escalates faster than density alone suggests; (2) the behavioural heuristics, not the contact forces, are what produce purposeful navigation; (3) bottleneck evacuation degrades beyond a critical crowd size, reproducing the faster-is-slower effect; (4) pressure is sharply localised at constrictions; and (5) the model reproduces the classical speed-density fundamental diagram as an emergent property. The simulator therefore captures the qualitative physics of real crowd disasters from simple local rules.

6.2 Further research

- Machine learning: replace the hand-tuned heuristics and force potentials with learned ones, following Deep Social Force, or use Social GAN-style trajectory prediction to calibrate parameters against real pedestrian data.
- Richer agent types: children, elderly, groups travelling together, and responders, expressed through the existing parameterisation.
- Mitigation studies: test how obstacles placed before an exit (a well-known counter-intuitive remedy), staggered release, or wider openings reduce peak pressure.
- Scalability: spatial partitioning to bring the quadratic neighbour search down and allow much larger crowds.

6.3 Ethical aspects

Crowd simulation is a dual-use tool. Used responsibly it improves the safety of venues, transport hubs and evacuation plans and can save lives. The same models, however, could in principle inform crowd-control tactics that endanger people, and calibrating models on real pedestrian footage raises privacy questions. Results must also be communicated with their limitations clear: a simulation is an approximation, and over-confident conclusions about a specific venue could give a false sense of safety. Our model is a research prototype and is not calibrated for certifying any real-world facility.

7. References

1. Helbing, D. & Molnar, P. (1998). Social Force Model for Pedestrian Dynamics. *Physical Review E*.
2. Moussaid, M., Helbing, D., & Theraulaz, G. How simple rules determine pedestrian behavior and crowd disasters. *PNAS*. (doi: 10.1073/pnas.1016507108)
3. Understanding Social-Force Model in Psychological Principles of Collective Behavior. *ResearchGate*.
4. Social Force Model for Pedestrian Dynamics. *ResearchGate*.
5. Gupta, A., Johnson, J., Fei-Fei, L., Savarese, S., & Alahi, A. (2018). Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks. *CVPR 2018*. (arXiv: 1803.10892)
6. Kreiss, S. (2021). Deep Social Force. *arXiv preprint*. (arXiv: 2109.12081)